

UNIT - II

MARKOV CHAIN / PROCESS

① Random Variable $\rightarrow X$

② Stochastic Process $\rightarrow$ The term 'Stochastic' means 'Random' & Process means 'variations of time'.

$\Rightarrow$ Markov Chain / Process $\rightarrow$ The outcome of given experiment can affect the outcome of next experiment. This type of process is called Markov process / chain.

Probability Vector $\rightarrow$ The n -dimensional row which contain non-negative values & their sum is always unity.

Ex:- $[0, 1, 0]$, $[0.7, 0.1, 0.2]$ ✓
 $[-1, 4, -3]$, $[0.5, -0.5, 0]$ ✗

Stochastic Matrix

$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$ here $[a_1, a_2, a_3]$, $[b_1, b_2, b_3]$, $[c_1, c_2, c_3]$ are probability vector.

Regular Stochastic Matrix

If the matrix P contain atleast 1 non-zero element so that P^n have non-zero values, then it is said to be a Regular Stochastic Matrix.

Sm. $A = \begin{bmatrix} 0 & 1 & 0 \\ 0.2 & 0.4 & 0.4 \\ 1 & 0 & 0 \end{bmatrix}$

Prob 2. Stochastic ✓
At least one non-zero ✓

$$A^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0.2 & 0.4 & 0.4 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0.2 & 0.4 & 0.4 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0.2 & 0.4 & 0.4 \\ 0.48 & 0.6 & 0.16 \\ 0 & 1 & 0 \end{bmatrix}$$

A^3

Q. $A = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 5 & 1 & -\frac{1}{2} \\ 0 & 4 & 0 \end{bmatrix}$

Check if Stochastic Matrix, Regular Stochastic Matrix or Not?

Solⁿ ~~Sum of 1st~~
 $[0, 0, 1]$, $[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}]$, $[0, 1, 0]$ are all probability vectors so it is like Stochastic Matrix

$$A^2 = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{8} & \frac{5}{16} & \frac{9}{16} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{8} & \frac{5}{16} & \frac{9}{16} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{14}{64} & \frac{22}{64} & \frac{10}{64} \\ \frac{5}{8} & \frac{2}{16} & \frac{1}{16} \end{bmatrix}$$

A^3 has all non-zero values
 it is Regular like Stochastic Matrix

$B \rightarrow [5, 1, -\frac{1}{2}]$ is not probability vector
So, it is not a Stochastic vector

Q. Check whether the vectors are P.V. or not

- i) $P = [\frac{1}{4}, 0, \frac{1}{4}, 0, \frac{1}{2}]$ Sum = 1 P.V. ✓
 ii) $P = [\frac{1}{6}, \frac{1}{3}, \frac{1}{2}]$ Sum = 1 P.V. ✓
 iii) $P = [\frac{1}{3}, \frac{1}{12}, \frac{1}{6}, \frac{5}{12}]$ Sum = 1 P.V. ✓

Transition Probability Matrix

$n=0$ initial	$n=1$ after 1 step			
$a \rightarrow b$	transition prob.	b_1	b_2	b_3
$P =$	a_1	P_{11}	P_{12}	P_{13}
	a_2	P_{21}	P_{22}	P_{23}
	a_3	P_{31}	P_{32}	P_{33}

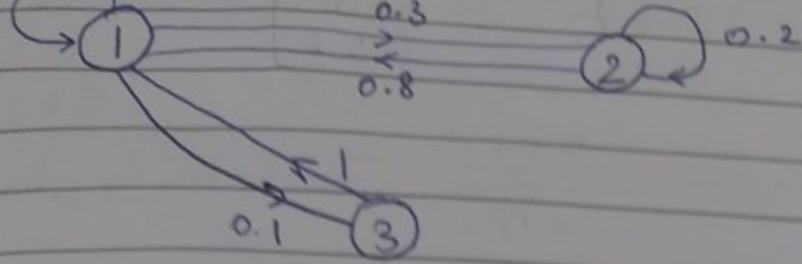
All values 0-1

Transition Diagram / Graph

Q. Draw transition graph for Markov chain w/ the following prob. transitⁿ matrix

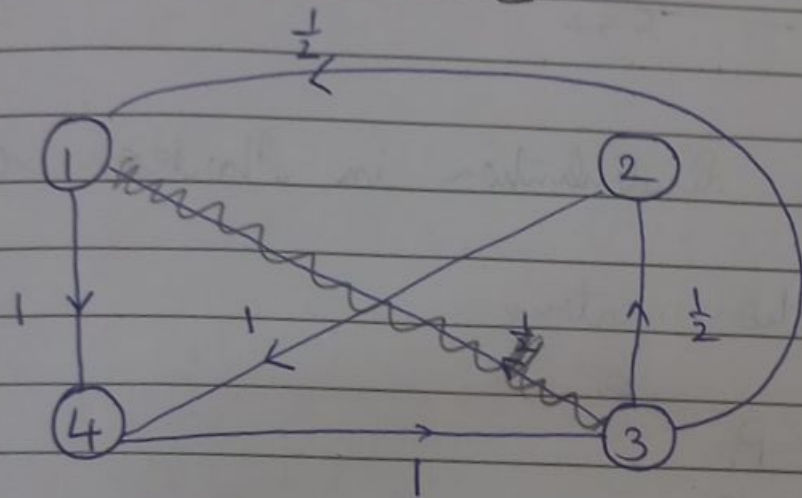
(1)

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.8 & 0.2 & 0 \\ 1 & 0 & 0 \end{bmatrix} \end{matrix}$$



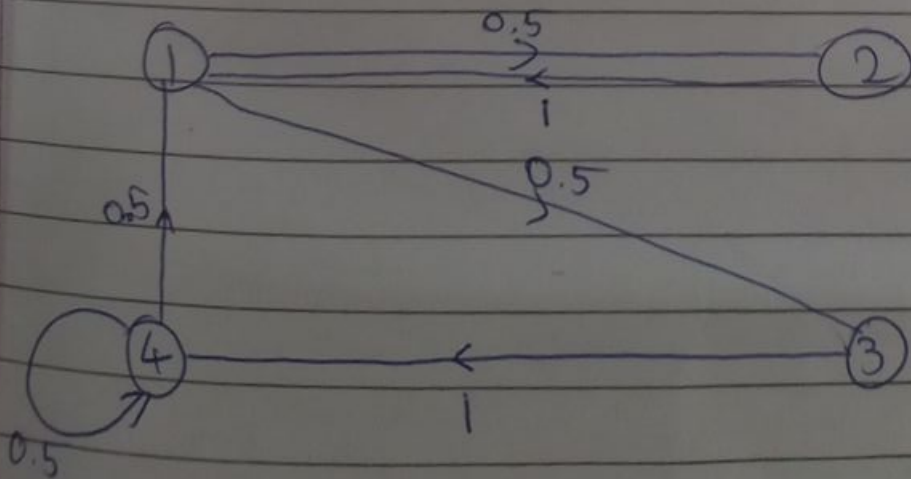
(ii)

	1	2	3	4
1	0	0	0	1
2	0	0	0	1
3	$\frac{1}{2}$	$\frac{1}{2}$	0	0
4	0	0	1	0



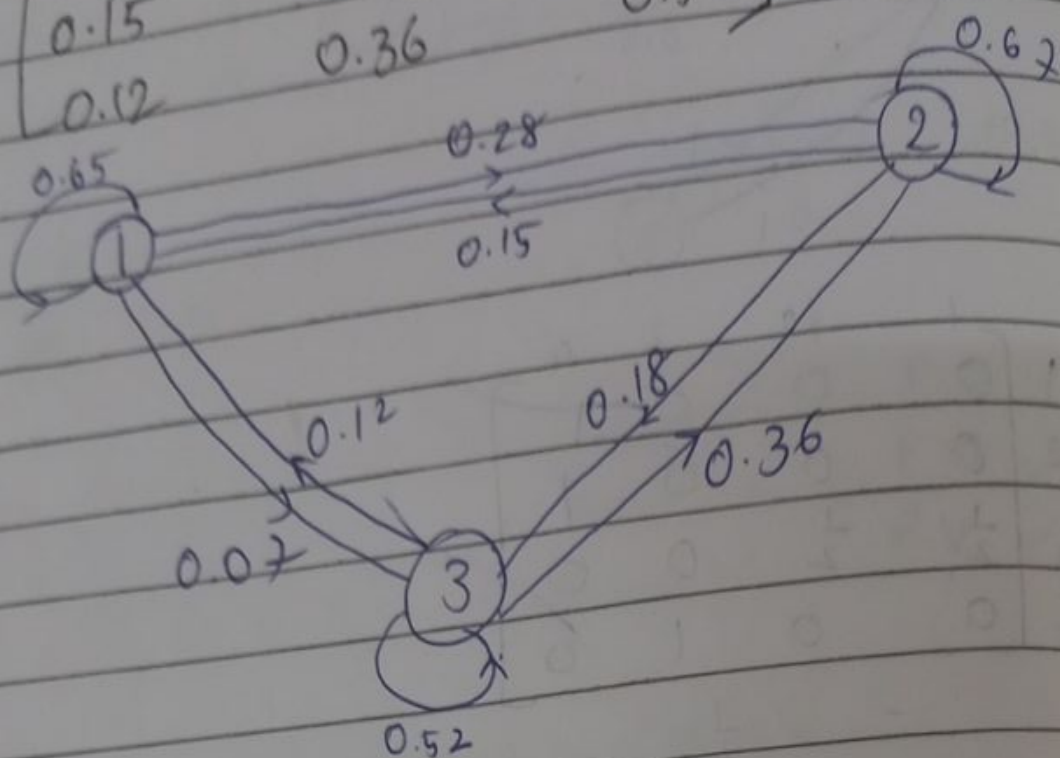
(iii)

	1	2	3	4
1	0	0.5	0.5	0
2	1	0	0	0
3	0	0	0	1
4	0.5	0	0	0.5



(iv)

	1	2	3
1	0.65	0.28	0.07
2	0.15	0.67	0.18
3	0.12	0.36	0.52



Probability Distribution in Markov Chain

$P \Rightarrow$ Transition matrix

$$P^1 = P^{(0)} \cdot P$$

$$P_{(1)} = P_{(0)} \cdot P$$

$$P_{(2)} = P_{(1)} \cdot P$$

$$P_{(2)} = P_{(1)} \cdot P^2$$

$$P_{(n)} = P_{(0)} \cdot P^n$$

On Sunday, $A \rightarrow B \rightarrow C$
 $V(0) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$

$n=0$

Transition Matrix, $P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix}$

Probability of C, Friday

$n=5$

$P(n=5) = V(0) \times P^5$

$= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix}^4$

$= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$

$\begin{bmatrix} \frac{2}{3} & 0 & \frac{1}{3} \\ \frac{2}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{2}{9} & \frac{2}{9} & \frac{1}{9} \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix}^3$

$= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$

$\begin{bmatrix} \frac{2}{9} & \frac{7}{9} & 0 \\ \frac{14}{27} & \frac{2}{9} & \frac{7}{27} \\ \frac{14}{27} & \frac{7}{27} & \frac{2}{9} \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix}^2$

$= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$

$\begin{bmatrix} \frac{14}{27} & \frac{2}{9} & \frac{7}{27} \\ \frac{14}{27} & \frac{2}{9} & \frac{7}{27} \\ \frac{26}{81} & \frac{59}{81} & \frac{2}{27} \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix}$

$=$

$= \begin{bmatrix} \frac{130}{243} & \frac{28}{81} & \frac{59}{243} \end{bmatrix}$
 A B C Ans

Q.2

~~B~~

$$P = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.1 & 0.7 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

Transition matrix P for the mobility in Village, Town, City.

Initial Proportion Vector, $P_0 = [0.7, 0.2, 0.1]$

After 1 year, $P_1 = P(0) \times P$

$$P_1 = [0.7 \quad 0.2 \quad 0.1] \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.1 & 0.7 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

$$P_1 = [0.38 \quad 0.39 \quad 0.23]$$

After 2 year, $P_2 = P(1) \times P$

$$P_2 = [0.38 \quad 0.39 \quad 0.23] \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.1 & 0.7 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

$$P_2 = \begin{bmatrix} 0.252 & 0.479 & 0.269 \end{bmatrix}$$

In the long run

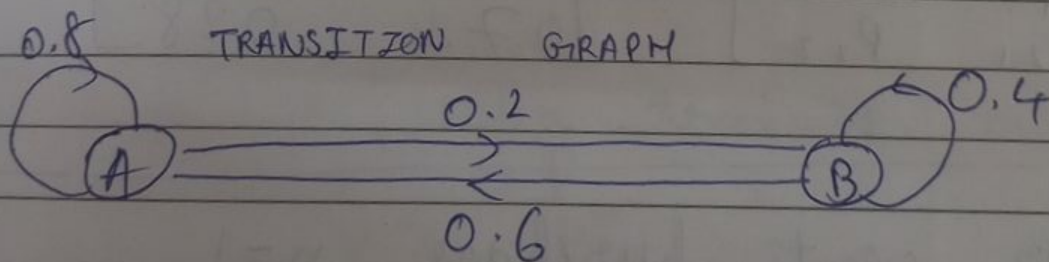
$$P(n) = P(0) \times P^n$$

2 manufacturers A & B

Solⁿ Initially, $V(0) = \begin{bmatrix} 0.8 & 0.2 \end{bmatrix}$

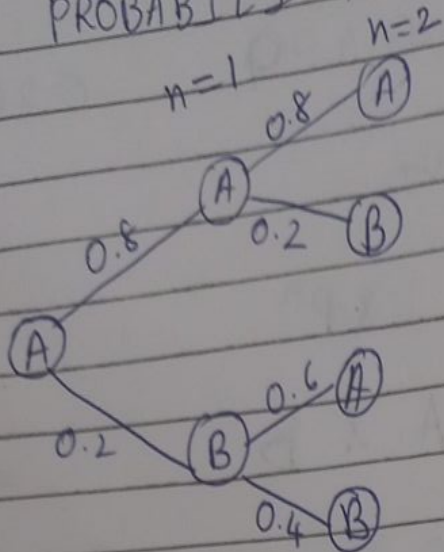
Present Purchase		A	B	Retention & gain
$n=0$,	$P =$	$\begin{bmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{bmatrix}$		$\downarrow$

Retention & loss $\rightarrow$



PROBABILITY TREE

Joint Probability



$A \rightarrow A$

$$P_{11} = 0.8 \times 0.8 + 0.2 \times 0.6$$

$$P_{11} = 0.76$$

$A \rightarrow B$

$$P_{12} = 0.8 \times 0.2 + 0.2 \times 0.4$$

$$P_{12} = 0.24$$

$B \rightarrow A$

$$P_{21} = 0.6 \times 0.8 + 0.4 \times 0.6$$

$$P_{21} = 0.72$$

$B \rightarrow B$

$$P_{22} = 0.6 \times 0.2 + 0.4 \times 0.4$$

$$P_{22} = 0.28$$

$$\begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} = \begin{bmatrix} 0.76 & 0.24 \\ 0.72 & 0.28 \end{bmatrix}$$

Your next purchase, $n=1$

Initial Prob. (Assumed) $\rightarrow V(0) = [1, 0]$

$$V(1) = V(0) \cdot P$$

$$= [1, 0] \begin{bmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{bmatrix}$$

$$V(1) = [0.8, 0.2]$$

$$V(2) = [0.8, 0.2] \quad \begin{bmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{bmatrix}$$

$$V(2) = [0.26, 0.24]$$

MULTI-STEP MARKOV CHAIN

The probability that a system which is in state a_i , will be in state a_j , after exactly 1 step is defined by P_{ij} .

We denote $P_{ij}^{(m)}$ probability which defines a_i will be in state a_j after exactly m steps.

It implies if $m=1$, $P_{ij}^{(1)} = P_{ij}$

$$P^{(m)} = \begin{bmatrix} P_{11}^{(m)} & P_{12}^{(m)} & P_{13}^{(m)} \\ P_{21}^{(m)} & P_{22}^{(m)} & P_{23}^{(m)} \\ P_{31}^{(m)} & P_{32}^{(m)} & P_{33}^{(m)} \end{bmatrix}$$

$P^{(m)}$ is m^{th} step transition matrix
 $P^{(1)} = P$
 $P^{(m)} = P^m$

Q. Transition matrix P of Markov chain is

$$P = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Initial prob. Distribution $P(0) = [\frac{2}{3}, 0, \frac{1}{3}]$

If $P_{ij}^{(m)}$ denotes m^{th} step transition probs
 Find $P_{32}^{(2)}$, $P_{13}^{(2)}$, $P^{(4)}$, $P_3^{(4)}$

$$P_{32}^{(2)} = 1^2 = 1$$

$$P^{(m)}$$

$$= P^m$$

$$P^2 = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

$$P_{32}^{(2)} = \frac{1}{2}$$

$$P_{13}^{(2)} = 0$$

$$P^{(4)} = P(0) \cdot P^4$$

$$P^4 = P^2 \times P^2$$

$$\frac{3}{16} + \frac{4}{16} + \frac{1}{16}$$

$$= \begin{bmatrix} \frac{1}{4} & \frac{3}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} \quad \begin{bmatrix} \frac{1}{4} & \frac{3}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} \frac{1}{4} & \frac{9}{16} & \frac{3}{16} \\ \frac{5}{16} & \frac{9}{16} & \frac{1}{8} \\ \frac{1}{4} & \frac{5}{8} & \frac{1}{8} \end{bmatrix}$$

$$P^{(4)} = \left[\frac{2}{3}, 0, \frac{1}{3} \right]$$

$$\begin{bmatrix} \frac{1}{4} & \frac{9}{16} & \frac{3}{16} \\ \frac{5}{16} & \frac{9}{16} & \frac{1}{8} \\ \frac{1}{4} & \frac{5}{8} & \frac{1}{8} \end{bmatrix}$$

$$P^{(4)} = \left[\frac{1}{4}, \frac{7}{12}, \frac{1}{6} \right]$$

$$P_3^{(4)} = \frac{1}{6}$$

Ans $\frac{3}{8}, \frac{1}{2}, \left(\frac{2}{16}, \frac{2}{16}, \frac{2}{16}\right), \frac{7}{16}$

Q. $P = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$

Initial Probability $= \left[\frac{1}{2}, \frac{1}{2}, 0 \right]$
Find $P_{13}^{(2)}, P_{23}^{(2)}, P^{(2)} = P_1^{(2)}$

Solⁿ $P^{(2)} = P^2$
 $P^{(2)} = \frac{1}{4} \begin{bmatrix} 2 & 0 & 2 \\ 4 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \times \frac{1}{4} \begin{bmatrix} 2 & 0 & 2 \\ 4 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

$P^{(2)} = \frac{1}{16} \begin{bmatrix} 6 & 4 & 6 \\ 8 & 0 & 8 \\ 11 & 2 & 3 \end{bmatrix}$

$P^{(2)} = \begin{bmatrix} \frac{3}{8} & \frac{1}{4} & \frac{3}{8} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{11}{16} & \frac{1}{8} & \frac{3}{16} \end{bmatrix}$

STEADY STATE or EQUILIBRIUM CONDITION

$$V = VP$$

$P \rightarrow$ Transition matrix

$V \rightarrow$ Steady state Probability / Fixed

Ex. $[x, y, z] = [x, y, z] \times P$ - (3) eqns.

$$x + y + z = 1$$
 - (4)

Q. A market analyst is interested in whether consumer prefer HCL or Wipro computers

Transition matrix	HCL	Wipro
$P =$ HCL	0.9	0.1
Wipro	0.35	0.65

$$V = VP$$

$$[x \ y] = [x \ y] \begin{bmatrix} 0.9 & 0.1 \\ 0.35 & 0.65 \end{bmatrix}$$

$$[x, y] = [0.9x + 0.35y, 0.1x + 0.65y]$$

$$0.9x + 0.35y = x$$

$$0.1x - 0.35y = 0$$
 - (1)

$$0.1x + 0.65y = y$$

$$0.1x - 0.35y = 0$$
 - (2)

$$x + y = 1$$
 - (3)

$$x = 0.78, y = 0.22$$

HCL $\Rightarrow$ 78%

WIPRO $\Rightarrow$ 22%

Q. Suppose that General Motors (GM), Ford (F) & Chrysler (C) each introduce a new car

	GM	F	C
P = GM	0.85	0.1	0.05
F	0.1	0.8	0.1
C	0.25	0.15	0.6

$$\begin{bmatrix} x & y & z \end{bmatrix} = \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 0.85 & 0.1 & 0.05 \\ 0.1 & 0.8 & 0.1 \\ 0.25 & 0.15 & 0.6 \end{bmatrix}$$

$$\begin{aligned} x &= 0.85x + 0.1y + 0.25z \\ y &= 0.1x + 0.8y + 0.15z \\ z &= 0.05x + 0.1y + 0.6z \\ x + y + z &= 1 \quad \text{--- (1)} \end{aligned}$$

$$\begin{aligned} -0.15x + 0.1y + 0.25z &= 0 \\ 0.1x - 0.2y + 0.15z &= 0 \\ 0.05x + 0.1y - 0.4z &= 0 \\ x + y + z &= 1 \end{aligned}$$

$x = 0.49$	GM $\Rightarrow$ 49%
$y = 0.36$	F $\Rightarrow$ 36%
$z = 0.15$	C $\Rightarrow$ 15%

Q.

$$\text{Transition Matrix } P = \begin{matrix} & \begin{matrix} A & B \end{matrix} \\ \begin{matrix} A \\ B \end{matrix} & \begin{bmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{bmatrix} \end{matrix}$$

Initially same market share initially, i.e., 50%.
Determine market share at steady state

Solⁿ

$$P(0) = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix}$$

$$P(1) = P(0) \cdot P = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{bmatrix}$$

$$P(1) = \begin{bmatrix} 0.7 & 0.3 \end{bmatrix}$$

$$P(2) = P(1) \cdot P = \begin{bmatrix} 0.7 & 0.3 \end{bmatrix} \begin{bmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{bmatrix}$$

$$P(2) = \begin{bmatrix} 0.78 & 0.22 \end{bmatrix}$$

$$P(3) = P(2) \cdot P = \begin{bmatrix} 0.78 & 0.22 \end{bmatrix} \begin{bmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{bmatrix}$$

$$P(3) = \begin{bmatrix} 0.812 & 0.188 \end{bmatrix}$$

$$P(4) = P(3) \cdot P = \begin{bmatrix} 0.812 & 0.188 \end{bmatrix} \begin{bmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{bmatrix}$$

$$P(4) = \begin{bmatrix} 0.8248 & 0.1752 \end{bmatrix}$$

$$P(5) = \begin{bmatrix} 0.8248 & 0.1752 \end{bmatrix} \begin{bmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{bmatrix}$$

$$P(5) = \begin{bmatrix} 0.82992 & 0.17008 \end{bmatrix}$$

$$A \Rightarrow 83\%$$

$$B \Rightarrow 17\%$$

Q.8 On January 1 (this year) Bakery A had 40% of its local market, B had 40%, C had 20%

Solⁿ ^{this year} $P(0) = \begin{bmatrix} 0.4 & 0.4 & 0.2 \end{bmatrix}$

Transition Matrix, $P = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{bmatrix} 0.9 & 0.05 & 0.05 \\ 0.05 & 0.85 & 0.1 \\ 0.1 & 0.07 & 0.83 \end{bmatrix} \end{matrix}$

Next Year / January, $P(1) = P(0) P$

$$P(1) = \begin{bmatrix} 0.4 & 0.4 & 0.2 \end{bmatrix} \begin{bmatrix} 0.9 & 0.05 & 0.05 \\ 0.05 & 0.85 & 0.1 \\ 0.1 & 0.07 & 0.83 \end{bmatrix}$$

$$P(1) = \begin{bmatrix} 0.40 & 0.374 & 0.226 \end{bmatrix}$$

$A \Rightarrow 40\%$

$B \Rightarrow 37.4\%$

$C \Rightarrow 22.6\%$

Steady State,

$$V = VP$$

$$\begin{bmatrix} A & B & C \end{bmatrix} = \begin{bmatrix} A & B & C \end{bmatrix} \begin{bmatrix} 0.9 & 0.05 & 0.05 \\ 0.05 & 0.85 & 0.1 \\ 0.1 & 0.07 & 0.83 \end{bmatrix}$$

$$-0.1A + 0.05B + 0.1C = 0$$

$$0.05A - 0.15B + 0.07C = 0$$

$$0.05A + 0.1B - 0.17C = 0$$

$$A + B + C = 1$$

$$A = 0.43$$

$$A = 74.3\%$$

$$B = 0.28$$

$$B = 72.8\%$$

$$C = 0.29$$

$$C = 72.9\%$$